



Contents lists available at ScienceDirect

Nuclear Inst. and Methods in Physics Research, A

journal homepage: www.elsevier.com/locate/nima

Muon imaging: Principles, technologies and applications

S. Procureur

Irfu, CEA, Université Paris-Saclay, 91191 Gif sur Yvette, France

ARTICLE INFO

Keywords:

Muography
Muon
Tomography
Detectors
Imaging

ABSTRACT

During the last 15 years muon-based imaging, or *muography*, has experienced an impressive development and has found applications in many different fields requiring penetrating probes. Structures of very different sizes and densities can be imaged thanks to the various implementations it offers: either in absorption/transmission or in deviation modes, not to mention the muon metrology for monitoring. The goal of this paper is to give an overview of the main principles of the muography, as well as the technologies employed nowadays and its current and potential applications. Considering the amount of studies dedicated to muography and the number of projects conducted in the last decade, this review focuses on the fields which are the most representative of the muography capabilities.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

The variety of radiations and particles offers nowadays numerous possibilities in imaging. The choice of the source depends on many parameters, in particular the nature of the object and its thickness. If several options exist below typically one meter, like X-rays or neutrons, deeper structures require very high energy and/or penetration capabilities which are often out of reach of artificial sources. For such cases, a particularly penetrating probe is offered by cosmic muons naturally produced in the atmosphere. Thanks to their high energy and large mass, they can indeed cross up to several hundreds of meters of rocks before being absorbed. Their potential for imaging has been known for decades, but the recent progress on particle detection driven by high energy physics has opened the door to a vast range of applications, resulting in a growing interest for this technique.

The next section is dedicated to the origin of these muons and to the principles of the different types of muography. The performance requirements as well as the most common detection techniques are described in Section 3, before reviewing some flagship applications in Section 4.

2. Cosmic rays and muography

2.1. Cosmic rays and muons

2.1.1. Primary particles

The extraterrestrial origin of this ionizing flux was conclusively discovered in 1912 with measurements under the sea and at high

altitude [1,2]. It took another few years to understand that the primary particles are charged, in particular through the measurements by Compton of the latitude effect [3] as well as the East-West effect predicted by Rossi [4–6]. Today we know that this particle flux, unfortunately called cosmic rays,¹ is composed of about 90% of protons and 9% of alphas, the rest being essentially electrons and some small fractions of heavier ions up to Fe. Though the solar wind gives a small contribution at low energy, most of the cosmic rays are produced outside the solar system, in particular in supernovae and in Active Galactic Nuclei (AGN). The intensity of cosmic ray nucleons shows a strong energy dependence approximately given by:

$$I_N(E) \approx 1.8 \times 10^4 \left(\frac{E}{\text{GeV}} \right)^\alpha \frac{\text{nucleons}}{\text{m}^2 \text{ s sr GeV}}, \quad (1)$$

with $\alpha = 2.7$ at energies below a few hundreds of TeV. In addition to the variations of the solar wind component, the intensity of the low energy part of the spectrum depends on the sun magnetic deflection and is thus anti-correlated with the solar activity (11-year cycle).

2.1.2. Decays in the atmosphere

When entering the Earth atmosphere, the primary particles interact with the atoms producing copious quantities of baryons and mesons. These particles can either reinteract or decay depending on their lifetime. In particular charged mesons may decay into muons, electrons

¹ This expression originates from Millikan, who was convinced that primary particles were photons.

E-mail address: Sebastien.Procureur@cea.fr.

<http://dx.doi.org/10.1016/j.nima.2017.08.004>

Received 30 June 2017; Received in revised form 26 July 2017; Accepted 2 August 2017
Available online xxxx

0168-9002/© 2017 Elsevier B.V. All rights reserved.

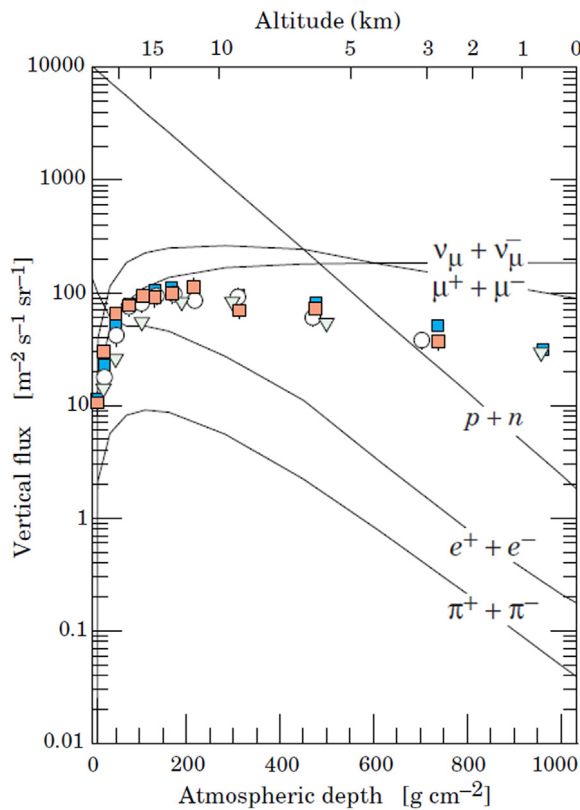
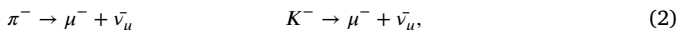


Fig. 1. Vertical fluxes of particles (except photons) originating from the cosmic rays as a function of the depth in the atmosphere, from [7]. Points are measurements for negative muons above 1 GeV.

and neutrinos, while neutral mesons quickly decay into photons. This cascade of reactions leads to the formation of a shower with a spatial extension reaching several hundreds of meters or more depending on the energy of the primary. The vertical flux of particles as a function of the atmospheric depth is shown in Fig. 1.

The two main sources of muon production correspond to the decay of charged pions and kaons:



with branching ratios of 99.99 and 63.5% respectively. It is worth mentioning that the decay into electron, which should be favored because of its lighter mass, is in this particular case strongly suppressed by an helicity effect, see e.g. [8].

In addition to this favorable production modes, muons have a significantly higher probability to survive the atmosphere crossing, benefitting from the combination of two effects:

- A longer mean lifetime ($\approx 2.2 \mu\text{s}$) compared to other unstable particles like mesons. Because of their high energy and the time dilatation from special relativity, this lifetime translates into a long mean decay length L in air given by $L \text{ (km)} = 6200 \times E \text{ (TeV)}$.
- A large mass compared to electrons ($m_\mu \approx 207 m_e$). The energy loss by Bremsstrahlung being proportional to $1/m^2$, it yields a value 40,000 smaller for muons compared to electrons at a given energy. The ionization and excitation therefore dominate the muon energy loss up to energies of the order of 500 GeV.

2.1.3. Flux at the sea level

As can be inferred from Fig. 1, the muon flux dominates by more than one order of magnitude the flux of other particles at the sea level.

It yields approximately $70 \text{ / (m}^2 \text{ s sr)}$ on the vertical axis, with a rough $\cos^2(\theta)$ dependence where θ is the angle to this axis. When θ increases, the larger muon path in the atmosphere favors the decay of low energy muons, resulting in a higher mean energy. A semi-empirical estimate of the mean flux at the sea level for high energy muons ($E > 100 \text{ GeV}/\cos\theta$) is given by:

$$\frac{dN}{dEd\Omega} = \frac{0.14E^{-2.7}}{\text{cm}^2 \text{ s sr GeV}} \times \left(\frac{1}{1 + \frac{1.1 \times E \cos\theta}{115 \text{ GeV}}} + \frac{0.054}{1 + \frac{1.1 \times E \cos\theta}{850 \text{ GeV}}} \right) \quad (3)$$

which shows the separate contributions from pions and kaons to the muon yield. At lower energies, the non-negligible muon decay probability as well as several atmospheric and solar effects complicate the determination of a mean muon flux, but systematic measurements and some parametrizations are available, see e.g. [9,10] respectively.

Though quite stable, the muon flux can be affected both in space and in time by several effects of different origins [11]:

- the altitude: as seen in Fig. 1, the muon flux increases above the sea level, essentially because of low energy muons which did not have time to decay;
- the latitude: the flux is lower close to the equator because of the shielding Earth's magnetic field;
- the solar activity (see previous section);
- the atmospheric pressure (short-time effect): a higher pressure in the lower part of the atmosphere results in a reduced flux, as the enhanced quantity of air absorbs low energy muons;
- the upper atmosphere temperature (seasonal effect): during summer, the increase of the upper atmosphere temperature enhances the mean free path of pions and kaons and therefore their probability to decay into muons [12].

Several simulation tools are nowadays available to study the precise development of cosmic showers in the atmosphere, like CORSIKA [13], and to efficiently propagate cosmic muons in thick matter, like MUSIC [14,15] or TIERRAS [16].

2.2. Muon imaging

As any other charged particles, muons interact with atoms of the matter they cross, resulting in a loss of energy and a change of direction (multiple scattering). These two effects underlie two different kinds of muon imaging as described in the next sections.

2.2.1. Transmission and absorption muography

Because of the energy loss, a muon has only a certain probability P to cross a given amount of material. If the mean energy loss (ionization, excitation and radiative) is large (resp. small) compared to the initial muon energy, P is very close to 0 (resp. 1).² The general equations for the mean energy loss (Bethe–Bloch and Bremsstrahlung) show that for a thin layer dx of material with density ρ , it primarily depends on the product $\rho \times dx$. At first order, the fraction of muons crossing (or non crossing) a material is therefore determined by the integrated density over the path length $\int \rho(x) dx$, a quantity called opacity. The experimental measurements of this fraction in different directions through an object from a given point of view (the position of the muon detector) then gives access to a cartography of the opacity, integrated along these directions. In the usual case where the thickness is actually known, the mean density is then obtained. This technique is commonly referred as transmission or absorption muography by analogy with the photography, though the term absorption is actually misleading.³ Though one or the other term is usually employed in a generic way, it should be emphasized

² The exact calculation of P is difficult, as it depends on the fluctuations around the mean energy loss, but it can be estimated through a simulation.

³ Indeed, muons either decay (μ^+ and μ^-) or are captured (μ^-) by the material.

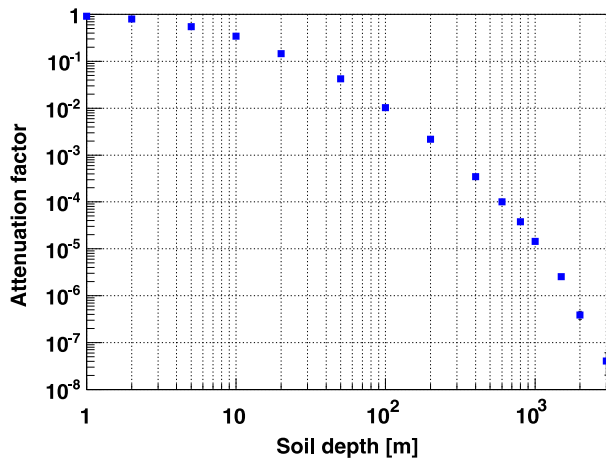


Fig. 2. Attenuation factor of the vertical cosmic muon flux as a function of the soil depth (density 2 g/cm³), simulated with the Geant4 package [17].

that they actually refer to distinct images, the transmission (resp. absorption) image being formed with crossing (resp. non-crossing) muons. The key ingredient of this technique is the broad energy distribution of cosmic muons, as some of them can cross several hundred meters of rock (see Fig. 2), which means that the opacity can still be measured for very thick objects.

By nature, the transmission muography measures the integrated opacity, and then provides 2D maps of an object. However, the combination of several 2D projections, either by moving the muon detector or by surrounding the object with several instruments, can give access to 3D information. As will be detailed in Section 3, a transmission muography apparatus consists in one or several detectors (telescope) which reconstruct the muon trajectory to infer its path through the object to image. In the transmission mode, the telescope is located downstream the object and reconstructs only crossing muons, while it is placed upstream in the absorption mode and complemented by an additional detector downstream acting as a veto to identify non-crossing muons. In both modes, the resulting muography image obtained from the accumulation of many muon tracks is then inherently blurred by two effects, namely the angular resolution of the telescope and the multiple scattering within the object and between the object and the telescope. An example of a raw transmission muography is given in Fig. 3.

2.2.2. Deviation muography

The multiple scattering originating from the collisions with the nuclei of a relatively thin material deflects the muon direction by an angle whose distribution is described by the Moliere theory, and approximately given by a Gaussian of zero mean value and standard deviation θ_{MS} :

$$\theta_{MS} = \frac{13.6 \text{ MeV}}{\beta c p} Q \sqrt{\frac{x}{X_0}} (1 + 0.038 \ln(x/X_0)) \quad (4)$$

where β is the relativistic factor, p the muon momentum in MeV/c, x the material thickness, Q the absolute electric charge of the muon and X_0 the radiation length empirically given by:

$$X_0 \approx \frac{716.4 \text{ g/cm}^2}{\rho} \frac{A}{Z(Z+1) \log(287/\sqrt{Z})}, \quad (5)$$

with Z and A the atomic and mass numbers. As for the muon absorption, the multiple scattering is thus sensitive to the opacity of the material, and to some extent to its atomic weight [19]. By surrounding an object with detectors reconstructing the upstream and downstream muon trajectories, one can then measure the diffusion angle.⁴ Assuming that a single main diffusion took place in the probed object,⁴ the diffusion

center can further be reconstructed in 3D, as the point of closest approach (POCA) between the upstream and downstream trajectories. The accumulation of muons then allows for the measurement of the typical diffusion angle as a function of the 3D position within the object, which can be inverted by dedicated algorithms to provide a 3D density map. More sophisticated, likelihood-based algorithms can also be employed assuming the muon did not scatter in the rest of the material, thus bringing information on the density of 3D voxels crossed by the muon [20]. An example of a deviation muography is given in Fig. 4.

Compared to the transmission mode, the deviation muography requires to install telescopes on both sides of the object, with one lower than the other, as illustrated in Fig. 5. The information collected in the deviation mode being richer, the imaging time in this mode is usually much reduced, as seen in Fig. 6. However, one should keep in mind that these two modes have in general different ranges of applications. The deviation technique is indeed only valid for modest opacity, as the diffusion center reconstruction is irrelevant when multiple diffusions occur.

2.2.3. Muon metrology

Though this is less directly related to imaging, it is worth mentioning here a third application of muons related to the metrology of structures. If muons indeed scatter in matter as discussed in the previous section, the mean scattering angle is still equal to zero. In other words, muons travel in average along straight lines. The detection of muons in several, distant instruments can thus supply their relative position and their potential movements with respect to each other [22,23]. A typical setup is presented in Fig. 7: a muon telescope is used as a reference and a remote detector is attached to the structure to monitor (in this case, a building). When a muon crosses both the telescope and the remote detector, the comparison between the extrapolated track from the telescope and the position measured remotely gives the position of the detector in the plane perpendicular to the muon track. The multiple scattering and the finite spatial resolution affect the measurement for individual tracks, but these errors decrease naturally with the accumulation of muons. This technique, which can be called *muon metrology*, is then equivalent to a natural, continuous laser positioning system, but works even in the presence of obstacles between the two instruments (wall, ground, etc.). Another advantage is that its precision is virtually infinite, as the error goes with $1/\sqrt{N}$ where N is the number of detected muons. Geant4 simulations showed that submillimetric resolutions can be obtained within one day with $50 \times 50 \text{ cm}^2$ instruments distant by 25 m.

3. Muon imaging technologies

The detection of charged particles like muons is a routine task in particle physics and has been achieved for decades in many accelerator facilities around the world. Cosmic muons themselves are studied since their discovery in 1936, and often used in laboratories for detectors characterization in cosmic test benches. Still, a given muography experiment has often to cope with several challenging – and at times contradictory – requirements from which the best technology should be determined. This is especially true for *in situ* transmission muography for which the environmental conditions may be somehow hostile.

3.1. Challenges of muography experiments

3.1.1. Detection area

The modest intensity of the cosmic muon flux obviously requires large size detectors to keep the acquisition time reasonable. Typical muography detectors have an active area above $30 \times 30 \text{ cm}^2$ and up to several m^2 , the final choice being a trade-off between the cost, the transportability of the instrument and the required statistics. It is worth mentioning that for large structures (e.g. a volcano) the real limitation is actually not the total number of crossing muons but rather their spatial dispersion.

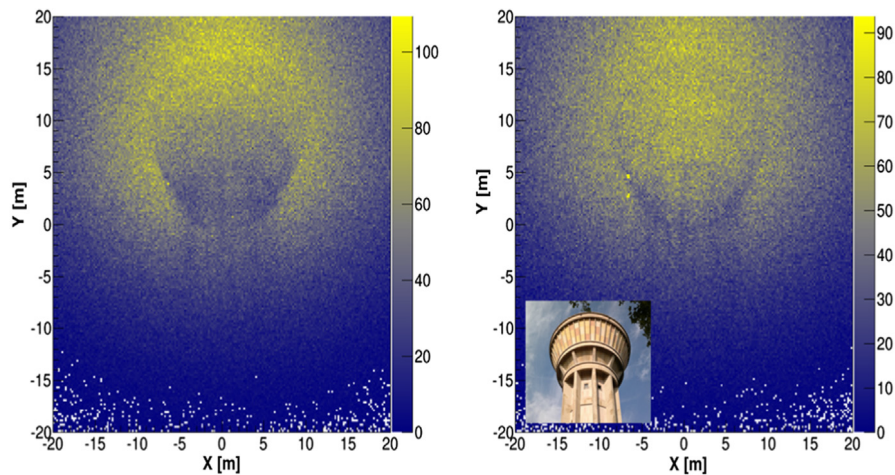


Fig. 3. Raw muographies of the Saclay water tower, with (left) and without (right) water in the tank. Each muography was obtained within 4 days of data taking using a small, 0.25 m^2 telescope [18].

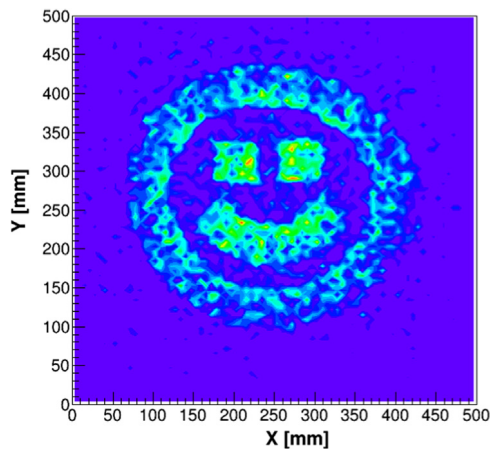


Fig. 4. Deviation muography of a cheerful arrangement of Lead bricks. The image shows the distribution of the POCA, weighted by the diffusion angle, projected in 2D in a plane perpendicular to the scanner axis [21].

3.1.2. Angular resolution

The most relevant parameter to characterize the accuracy of a muography instrument is the angular resolution. The optimal value strongly depends on the application, and can range from sub-mrad for low Z detection in deviation mode to roughly 10 mrad for large structures in transmission. In the latter case, the appropriate value is driven by the spatial extension of the heterogeneities one wants to detect, the overall accuracy being intrinsically limited by the amplitude of the multiple scattering within the object. This limitation explains why detectors with modest resolution (around 1 cm) can still be a valid option in some transmission applications, even if a better resolution directly improves the overall compactness and the opening angle of the apparatus. Obviously the resolution requirement is often in contradiction with the need for large area detector, as it implies a very large number of electronic channels. A way to overcome this issue is to design detectors whose resolution is much below the standard $p/\sqrt{(12)}$ value,⁵ as can be achieved with drift chambers or with multiplexing.

3.1.3. Acceptance

In the case of big structures, the instrument should have a wide opening angle to probe a large fraction of the object. It may also

be interesting to have some open-sky flux within the acceptance, for example to monitor the flux variations (for dynamical measurements), or to better normalize the data with a simulation to extract the mean density. Compact instruments are then preferable, though muon telescopes are by nature quite directional. An interesting alternative is provided by instruments with a 2π solid angle which can thus image all the surroundings simultaneously. This is particularly attractive for underground measurements like in boreholes or in sites with several structures to image.

3.1.4. Robustness

For a wide class of *in situ* applications (volcanoes, pyramids, boreholes, etc.), the instrument should be transported in harsh conditions, in the air and/or on trails. Vibrations can easily damage a detector component (like glass) or move a piece of dust inside the active volume, not to mention the unsoldering or shock on the electronics or the embarked PC.

Once installed, the apparatus should further resist to the environmental changes, like humidity, dust or storms. The temperature variations can also induce a change in the detection efficiency, or progressive unscrewing of some components. These conditions differ drastically from the controlled atmosphere of experimental halls. As explained in Section 4, several telescopes have already been operated in such conditions over the last few years, proving that carefully designed instruments can indeed be used in a tough environment.

3.1.5. Autonomy and access

A typical transmission muography experiment routinely lasts several months. With limited, or even absent infrastructure nearby, the apparatus should run on its own. In particular, the electric consumption needed for all the tasks – high voltages, electronics supply, PC for monitoring, acquisition, data storage – should be minimized, to fit within local energy production like solar boards. Standard telescopes consume a few tens of Watts, *i.e.* the equivalent of a bulb. For gaseous detectors, gas leakage and outgassing should be kept small to limit the storage of bottles. If a quasi-online data analysis is required, a network access should also be installed to transfer the data. This remote access also permits regular checking and monitoring, with the possibility of distant intervention to fix a problem (acquisition crash, software bug, high voltage tuning, etc.). In the extreme case where no energy is available, the energy consumption can be lowered to zero with passive detectors like emulsions.

⁵ Where p refers to the pitch of the detector, *i.e.* the distance between two readout elements.

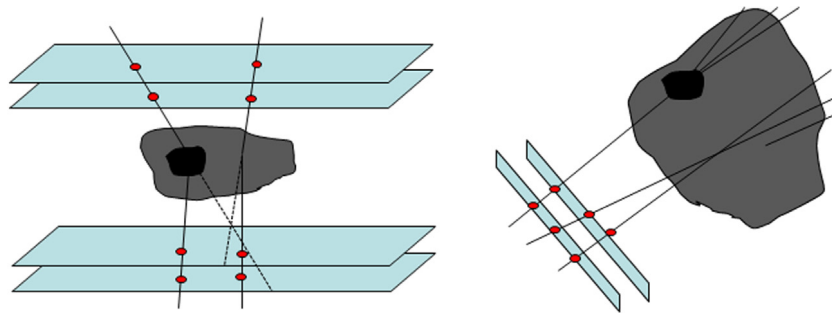


Fig. 5. Principle of the deviation (left) and transmission (right) muography. Zones of overdensities are represented in black within the gray objects.

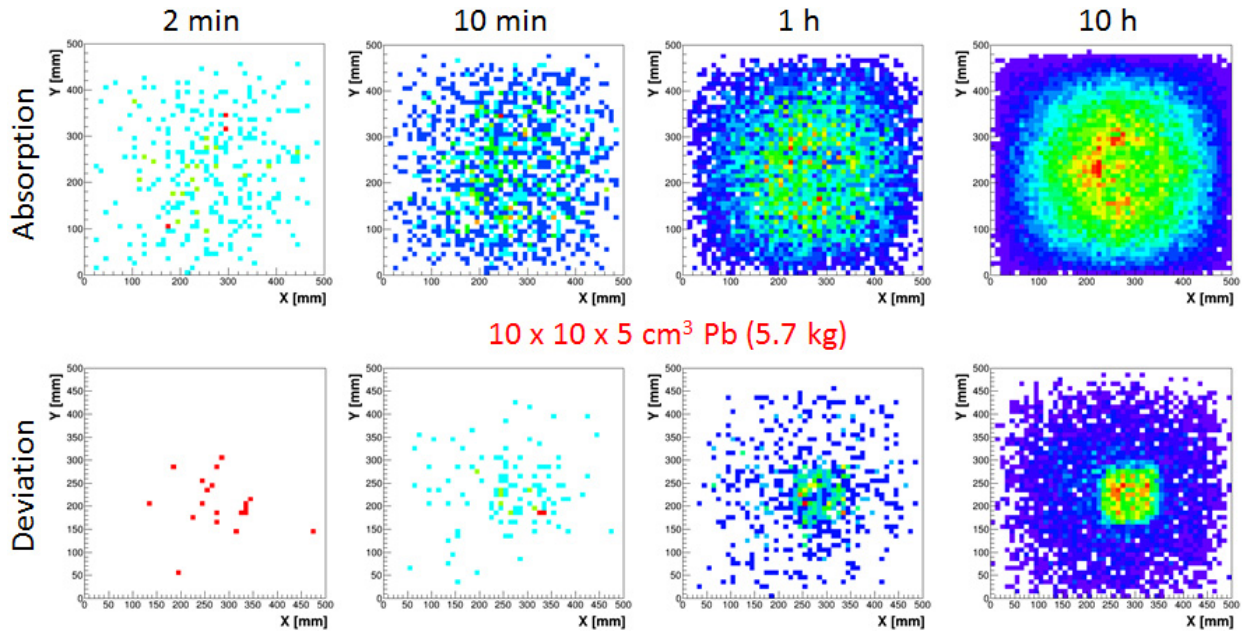


Fig. 6. Transmission (top) and deviation (bottom) muography of a single Lead brick obtained from the same data, as a function of the acquisition time. The reconstruction of the brick in deviation is much faster, as the deviation mode uses more information.

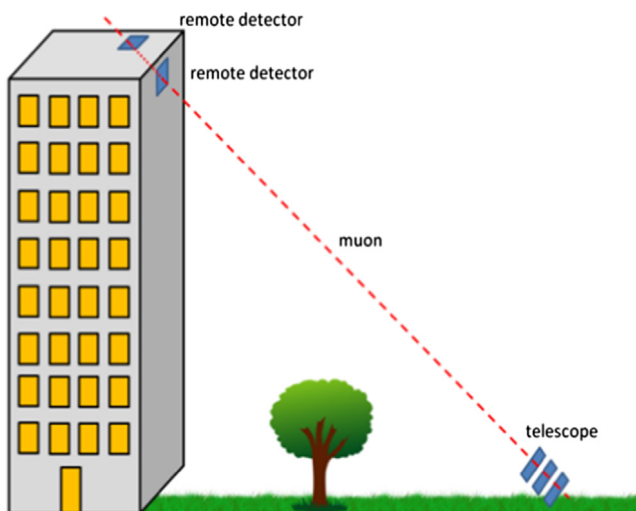


Fig. 7. Principle of the muon metrology: the simultaneous reconstruction of muon trajectory in a reference telescope and the position in a remote detector attached to a given structure provides the relative position of the remote detector.

3.1.6. Miscellaneous

In addition to the above specifications, other parameters may be relevant for a given application. For example, when muons can cross the telescope from both sides (like in volcanology or horizontal measurements), the time resolution of the detector is a key issue to separate both contributions. Beyond the performance itself, the cost and the industrial production capability are also of particular importance for applications aiming at massive deployment like homeland security or civil engineering. This generally calls for a phase of know-how transfer to the industry which can be highly non-trivial if the manufacturing process is too complex, or if the market potential is not convincing enough for the industrial.

3.2. Detection techniques

As already stated, some of the criteria described above are contradictory. Choices have therefore to be made, and historically two classes of instruments appeared – scintillators and nuclear emulsions – each focusing on different requirements. With the emergence of deviation muography, as well as the progress on instrumentation, gaseous detectors showed up as a third, intermediate option in the 2010s.

3.2.1. Plastic scintillators

Scintillators use the light created by the excitation–deexcitation of atoms interacting with the incoming muons. Photons are collected,

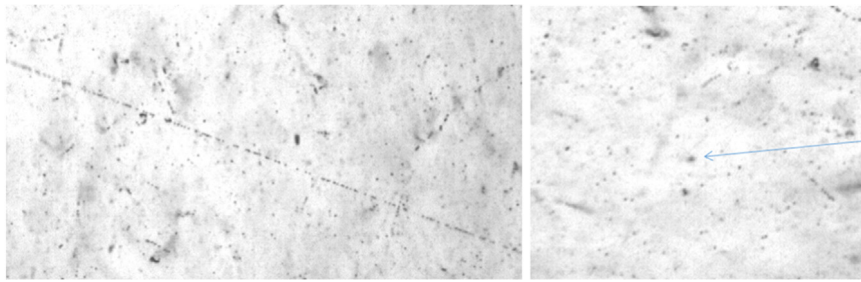


Fig. 8. (Left): Cosmic ray muon in the plane of an emulsion film as seen in transmitted light in an optical microscope. The track is visible as a sequence of tightly aligned grains after development (usually better than 50 nm). The average size of grains is about 0.5 μm (Right): Cosmic ray muon crossing the plane of an emulsion film nearly orthogonally, i.e. a more standard case in muography. The track is here visible as a single grain after development in the focal plane of the microscope. As the film thickness spans about 80 μm and the field depth is about 5 microns by changing the focal plane different grains of the same track can be seen. The grain appears larger than others because the shadows of other grains of the same track but out of the focal plane appear blurred and superimposed. Courtesy of C. Bozza.

converted into electrons which are then amplified in a photo-multiplier or a Silicon PM. Scintillators can be optically coupled with WLS fibers which absorb the primary light and re-emit photons at more appropriate wavelengths for the PM.

Scintillation detectors have been used for more than a century (see e.g. the spinthariscopes), and combine several advantages for muography. They are indeed very robust, easy to build, relatively cheap, practically insensitive to environmental conditions,⁶ and provide detection efficiency close to 100%. Their main drawback is their spatial resolution $p/\sqrt{12}$ which is determined by the width p of the slat. With typical slat ranging from a few mm to a few cm, the angular resolution can be kept within 3 to 10 mrad with meter-like lever arms.

3.2.2. Nuclear emulsions

Another technique routinely used in muography is the nuclear emulsion plate. The principle of this detector is the same as the analog photography: silver halide crystals are mixed in gelatin and coated on a glass substrate. The interaction of muons first reduces the silver ions into metallic silver. After the exposure, the emulsion is then developed to force the reduction reaction around the sites already reduced. In the last step, the remaining silver halides are dissolved (fixation), leaving a plate darkened at the muon positions, see Fig. 8. Compared to standard photography, the nuclear emulsions are thicker to reconstruct the muon trajectory in 3D and with much higher granularity, resulting in sub-micron resolution. Such a resolution goes beyond the blurring due to the multiple scattering within the object, but allows for extremely compact instruments and large acceptance. The track pattern analysis and the alternance of emulsions with Lead or Iron plates can also help to disentangle high energy muons from low energy ones or electrons. Moreover, this instrument is fully passive and then ideal when no energy source is available. However, the emulsion lifetime is often limited to a few months and further drops with high humidity or with temperatures above roughly 25 $^{\circ}\text{C}$. Furthermore, the data analysis can only take place after the exposure and is preceded by a time consuming, automatic scanning to reconstruct the tracks and form the final image. State-of-the-art scanners like the Hyper Track Selector of the Nagoya University can now process several square meters a day. Last but not least, no time information is recorded by the plates which only accumulate the muons during the whole exposure, preventing from any dynamical monitoring of the structure.

3.2.3. Gaseous detectors

The muon detection relies here on the interaction with a gas mixture leading to ionizations along the muon trajectory [24]. Under proper electric fields, electrons drift towards an anode (plane or wires) and are accelerated to further ionize the gas, as shown in Fig. 9. A charge avalanche then develops in the vicinity of the anode, and the movement

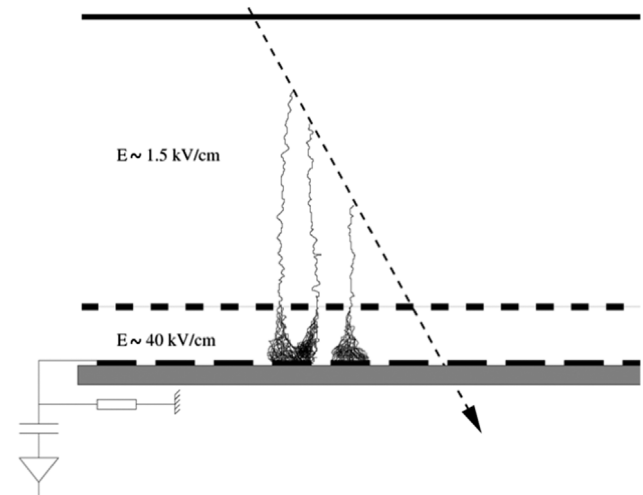


Fig. 9. Detection principle of a gaseous detector, the Micromegas. An incident particle ionizes the gas in the conversion gap. Electrons are then sucked through a micro-mesh to a small amplification gap in which they create charge avalanche under a high electric field.

of the electron-ion pairs induce a charge collected on readout elements (wires, strips, pixels) connected to a dedicated electronics.

Numerous advantages triggered the development of gaseous detectors for muography projects. In particular, they can achieve resolutions of a few hundreds of microns, i.e. at least of factor of 10 better than scintillators, at still a reasonable cost. Unlike emulsions they also allow for dynamical studies, thus combining the advantages of the two historical technologies. Various solutions also exist to limit the number of electronic channels:

- thanks to the signal time measurement, drift-based chambers offer resolutions at least 3 times better than $p/\sqrt{12}$;
- the muon signal on Micro-Pattern Gaseous Detectors (MPGDs) being shared on several strips, multiplexing patterns have been implemented to reduce by a factor of ~ 15 the number of channels while keeping a high detector granularity [25].

The amplification of the charges being highly dependent on the gas temperature and pressure, environmental changes affect the gain and degrade the overall performance — with a risk of saturation or efficiency loss. An online feedback on high voltages then prevents from large oscillations of the gain for *in situ* operations.

Last but not least, several Time Projection Chambers (TPC) are currently being developed, and may represent the ultimate instrument for muography. By measuring a 3D portion of a track, they offer better compactness for the same resolution. More importantly, they can detect

⁶ The sensitivity rather originates from other components like the SiPM for the temperature.

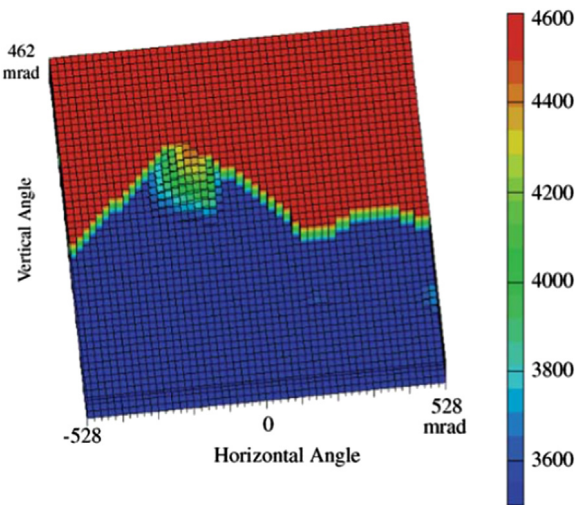


Fig. 10. Transmission muography of the Mt. Asama obtained by the difference between measurements and simulation. The crater near the top clearly appears as a lower density. Source: Plot from [29].

muons with 2π coverage and practically isotropic resolution. Such instruments are ideal candidates for borehole measurements where the space is highly constrained, and more generally for the scanning of underground and large structures. Other interesting features include the possibility to reject background like electrons or low energy muons, and the online measurement of the complete open-sky (if any) which can be highly valuable for simulation comparison and flux subtraction.

4. Current and future applications

Driven by the deep penetrating power of muons and the non destructive feature of this technique, many proposals for applications have emerged in the last two decades. Schematically, muons can go wherever there is no physical access, or potentially too many hazards. This section aims at presenting a non-exhaustive overview of current projects as well as a glimpse of the fields in which muon potential is already acknowledged.

4.1. Volcanology

After a couple of historic experiments in the XXth century [26,27], the revival of muography was initiated in Japan for the study of active volcanoes. After Nagamine's original idea in 1995 [28], Tanaka performed the first muon imaging of a volcano in which he discerned the central crater [29], see Fig. 10. This pioneer experiment convincingly showed that muography can complement traditional measurements using micro-gravimetry or resistivity.

In France, the Diaphane collaboration started in 2007 has built several scintillator-based telescopes and made major contributions in the field. They performed original methodological developments, setting in particular feasibility formulas for muography analysis [30]. During long and continuous measurements campaigns on several volcanoes (Italy, Philippines, France), they optimized the telescopes with a third layer of scintillator, to suppress the non negligible fraction of fake tracks coming from cosmic showers and noise [31]. Later they discovered, and corrected, a bias in the density estimate from muography images due to the presence of an upward muon flux in some configurations (like the presence of a long valley behind the telescope) [32]. In 2014, Tanaka's team and Diaphane independently performed the first joint inversions of data from gravimetry and muography [33,34], quantifying the benefit

of each sets of data.⁷ The Diaphane study was recently complemented by the first 3D joined inversion between muography (3 independent detectors) and gravimetry data [35]. Last but not least both teams recently presented interesting evidence for a variation of the muon flux in some regions of their respective volcanoes, temporally correlated with recorded activities near the surface (fumarole appearance and magma movement). Such observations fully justify the hope that permanent monitoring of active volcanoes with muon instruments may be one day in position not only to better understand their structure but also to alert the surrounding population of a risk of eruption. Fig. 11 displays the results of a Principal Components Analysis (PCA) performed over several months from a single muon detector on the slope of La Soufriere de Guadeloupe. The time distribution plot reflects the dynamics of the inner hydrothermal system where large volumes exchange heat and water content [36].

In view of the great scientific and societal potential, other collaborations were formed involving physicists and geologists to equip more volcanoes with muon instruments. The TOMUVOL collaboration has built a large, 1 m^2 RPC-based telescope whose performance has been thoroughly estimated on the Puy de Dome during the last years [37]. The structure of this ancient volcano is very well known, and thus constitutes an ideal showcase to test the density determination and to benchmark the different technologies [38]. Common measurement campaign already took place with the MU-RAY telescope [39,40], and detailed simulations were developed to better understand the different sources of background.

Several world class instruments also emerged in Italy which hosts a very large fraction of active volcanoes in Europe. The MU-RAY scintillator telescope from an INFN/INGV collaboration was built in 2009 and featured a special arrangement of scintillator bars to improve its angular resolution [41]. After a first successful campaign on the Vesuvius in Italy, the first laboratory dedicated to muon measurements on this volcano is currently in preparation, with an improved telescope including better background suppression capabilities. The Stromboli [42] and Etna volcanoes are also the subject of different muon studies, the latter with an innovative Cerenkov detector. With the multiplication of such projects, and the participation of more institutes e.g. from Belgium and Hungary, discussions are underway to better organize and structure these activities at an international level. The Japanese ERI already signed several MoUs with European institutes, and a proposal of a European network dedicated on muography has been submitted, mainly driven by the volcanology community.

4.2. Archeology

The non-invasive feature of muography has naturally elicited the interest of archeologists. A pioneer experiment was conducted in the late 1960s by Luis Alvarez [27], whose team installed large area spark detectors in the Belzoni chamber of the Chephren pyramid. Facing against large noise in the detectors and additional unusual difficulties,⁸ the measurement concluded to the absence of hidden rooms of significant size.

Because of their size, pyramids remained as a privileged field of investigation for muons. In 2011, Arturo Menchaca and his team installed Multi-Wire Proportional Chambers in the Mexican Pyramid of the Sun at Teotihuacan [44]. After three years of data taking, they successfully reconstructed the shape of the pyramid, in spite of its large inhomogeneities.

⁷ In a nutshell, muography measurements bring constraints on the upper density of the volcano, and help the gravimetry to gain precision on deeper regions inaccessible by muons.

⁸ It is enough to quote Alvarez memories to have a glimpse at the troubles the team had to face [43]: "The equipment first operated on June 5 [1967]. The next day the Egyptian-Israeli Six-Day War broke out. Early that June morning Israeli aircraft successfully avoided Egyptian radar and destroyed almost the entire Egyptian air force on the ground [...]. Egypt accused the United States of helping Israel, broke off diplomatic relations, and interned our foreign-service officers".

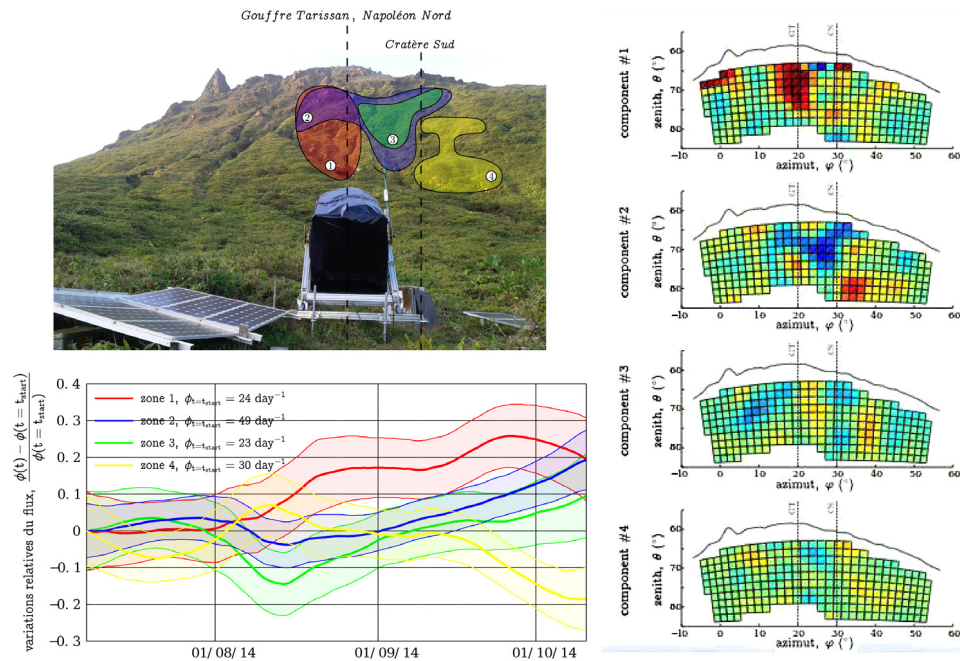


Fig. 11. Functional imaging of the Soufriere of Guadeloupe dome. Left top : picture of the muon detector facing the dome. Projected on the surface of the dome, the 4 different areas with coherent time variations (left, bottom). Right : principal vector components analysis of the 4 areas.

In 2015 another muography project was launched by the HIP institute and the University of Engineering in Cairo, the ScanPyramids mission [45]. A team from the Nagoya University first installed several m² of emulsion plates in the Bent pyramid at Daschour, and convincingly observed the known upper Chamber. Starting from 2016, new emulsions have been installed in various places of the Khufu pyramid at Giza, and detected a void behind the Chevrons of the North face [46]. Plates are associated in doublets and rotated before and after the installation to discriminate muons impinging the plates off the exposure duration. After developments in Cairo, the emulsions are analyzed in the Hyper Track Selector which can process 0.5 m² of plates every hour. A 1 m² scintillator telescope from the KEK has also been installed in the Queen Chamber with thin, 1 cm slats to complement emulsion measurements. Three Micromegas-based telescopes of 0.25 m² each were further built by CEA and positioned outside the pyramid, aiming at scanning its edges. These gaseous instruments currently face the extreme conditions of the Egyptian desert: temperature variations of more than 30 °C, dust, sand storms, grounding issues, etc. After three months of data taking, an already known cavity was observed (C2) and another one was reported at roughly 110 m high (C1), see Fig. 12. The mission is currently accumulating more data to scan other zones of the Great Pyramid.

Pyramids are not the only archeological targets which can be probed by muons. The deep inside structure of other large man-made constructions can also be accessed, and a project is underway to scan the Kastasi Amfipoli Macedonian Tumulus in Greece [47]. The detection of large cavities in mountains could also reveal the presence of ancient caves, particularly around existing sites where suspicions of unknown voids exist (Lascaux, Tharros, Qumran, etc.). The deviation muography has also been recognized as a powerful tool to investigate smaller objects like statues hosting metallic structures whose shape and state can help to determine their origin and potential need for restoration. Finally, in addition to the standard muography, the muon metrology technique has been considered to monitor the movement and deformations of old buildings, like the Palazzo della Loggia in Brescia [23].

4.3. Homeland security

The original work of the Los Alamos team in 2003 [19] on deviation muography showed direct applications for homeland security, and more

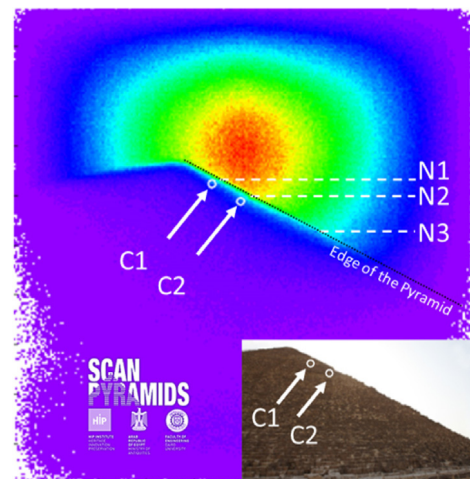


Fig. 12. Result of the Great Pyramid muography obtained from the East face, with the position of the external notches (N1, N2, N3) and the two cavities (C1, C2) detected from the analysis of muon distribution along the edges [46].

particularly the detection of the so-called Special Nuclear Material (SNM) like contraband Uranium in cargos and containers. Conventional searches of SNM include artificial radiation – X-rays, neutrons – which suffer from several drawbacks:

- the installation cost, including the material (source) and the safety;
- the radiation risks around the source, including in the container (potential presence of migrants);
- the possibility to hide SNM with various shieldings like Lead.

The ability of muons to detect the shielding itself as well as their sensitivity to the density and to the Z of the material then appeared as an elegant alternative to artificial radiation scans. In the US, a collaboration between Los Alamos and the Decision Sciences company resulted in the installation in 2012 of the first detection device in the world based on

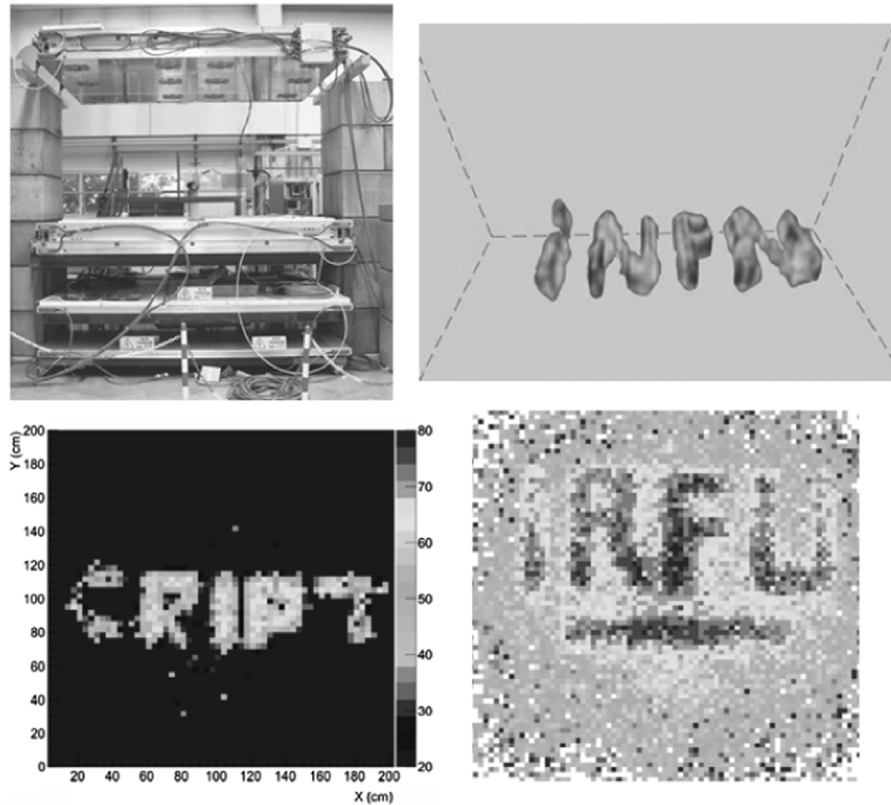


Fig. 13. (Top left): one of the largest muon scanners in the world, at INFN Legnaro, and images obtained with Lead bricks using the Legnaro (top right), CRIPT (bottom left) and Irfu (bottom right) scanners.

deviation muography [48]. It seems, however, that the difficulty to manufacture the detectors (drift tubes) has so far prevented this company from building other such devices in the last years, though a contract was recently awarded with AWE. Similar technologies are under study in several countries, in particular in Italy (INFN/Legnaro [49]) where a very large, 11 m^3 muon scanner built from CMS Drift Chambers is in operation since 2009, see Fig. 13 (top). On the contrary, groups in Canada (AECL/Carleton [50], see Fig. 13 (bottom left)) and Italy (INFN/Catania [51]) have opted for easier-to-build, scintillator-based technologies, at a cost of slightly less compact scanners. MPGDs and RPCs are also investigated in the US (FSU, GEM detectors [52]) France (Saclay, Micromegas [21], see Fig. 13 (bottom right)), Great Britain (Bristol, RPC [53]) and Israel (Lingacom Ltd, THGEM). In order to be viable, such scanners should ensure the detection of the equivalent of a few kg of Uranium in typically less than 1–2 min, as defined by the US Domestic Nuclear Detection Office (DNDO) [54]. This detection time is primarily determined by the angular resolution of the setup, but can be further reduced by estimating the (low energy) muon momentum, using either magnetic deflection or time of flight measurements for very precise detectors like scintillators.

4.4. Nuclear reactor and waste imaging

Beyond homeland security, the muon sensitivity to high-Z material has been investigated for already several years for the characterization of nuclear waste. Since the 1950s, spent nuclear fuel is essentially stored in dedicated pools or in Dry Storage Containers (DSC). The latter generally consists in a cask in which the nuclear material is inserted, for example in the form of bundles or pencils, and sealed for long term storage. To minimize the outside radiation level as well as the risk of leakage, various concrete and/or metallic shieldings surround the fuel, thus largely preventing from non-destructive inspection with gammas or neutrons. The need for later inspection of DSCs is however critical

to monitor the long term stability of the cask, to anticipate any risk of fuel leakage, and also to characterize old containers for which a precise and systematic description of the stored material may be absent. If this application requires (much) finer imaging than for the homeland security, the detection time is not as stringent and measurements of several days or weeks are truly acceptable. In some cases, permanent muon imaging devices directly integrated to the DSC could be designed for very long term monitoring in an autonomous way.

The imaging performance of such setups has been demonstrated through numerous simulations, see e.g. [55–59]. If no explicit degradation of the vessel itself has been tested yet (associated to a risk of leakage), the nuclear waste content of a cask can be accessed with great precision. Fig. 14 shows in particular a simulation of a cask containing several full and empty fuel assembly compartments imaged with a cylindrical detector. Empty compartments clearly show up at a 18 sigma level with a muon statistics equivalent to approximately 2 days, thus reaching a 5% accuracy on the compartment filling.

Interestingly, if deviation muography obviously yields a much better imaging of the container, absorption muography is still competitive due to the relaxed imaging time. This method might even exceed the precision of the deviation mode in the case of large and dense containers, as it is much less affected by the blurring of multiple diffusions.

4.5. Other applications

Though the topics described above have gathered the main part of the community, the muography potential has been acknowledged or demonstrated in many other fields. As already stated, an exhaustive list is beyond the scope of this review. Some of these fields are however worth mentioning, as they could nurture future trends in muography.

4.5.1. Underground measurements

The exploration of soils and underground is a vast but natural domain for penetrating muons. Though the technique is non destructive,

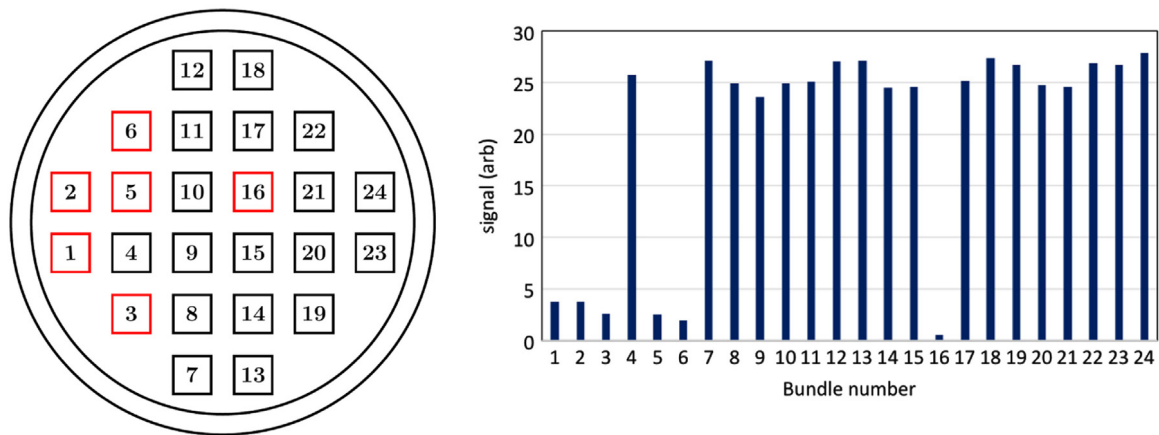


Fig. 14. (Left): geometry of a simulated cask containing full and empty (in red) fuel compartments. (Right): corresponding integrated areal density (related to the radiation length) of each compartment after 1.9 days of statistics.

Source: Plot from [59].

it requires in this case an access below the structures to observe (mines, gallery, boreholes, etc.). Several experiments have already been conducted for the localization of ore in existing mines, see e.g. [60] with the aim to facilitate the detection of heavy materials like Uranium, Gold or Lead. Muography instruments could in particular be deployed within boreholes during the search in new ore sites, to reduce the number of drillings and thus the costs of the exploration. Such instruments have to cope with severe constraints like very limited space and potential vibrations or shocks during the insertion. 3D mapping of a whole site can be achieved through the deployment of an array of such instruments (like TPCs) in several neighboring boreholes, though the proof of concept is still to be done. Beyond ore detection, the whole field of geoscience could benefit from this new class of underground instruments, to study the hydrology of soils, geological shells, faults, or perhaps to monitor one day deep storage sites like for CO₂ [61].

4.5.2. Nuclear reactor imaging

The nuclear disaster occurred in 2011 at the Fukushima-Daiichi nuclear plant after a tsunami in Japan has revealed the possibility to probe nuclear reactors and check their integrity [62]. After preliminary tests in the US [63], a first measurement campaign took place at Fukushima in 2015 using a transmission setup. In spite of very limited contrast, the results suggest that the molten fuel has disappeared from the primary containment vessel [64]. Another experiment has been announced, based this time on larger detector working in deviation mode to achieve better precision. While such an experiment will hopefully remains unique, muon investigation of old nuclear plants could help to check their integrity prior to any decommissioning.

4.5.3. Civil engineering

More prospective applications are very recently emerging in the field of civil engineering, especially in urban environment. The stability of buildings or the development of a cave-in can be monitored while tunnel boring. The imaging of supporting pillars of various ancient structures like bridges could also help to anticipate their need for restoration.

5. Conclusion

Within fifteen years, muon-based imaging has become a mature technique routinely used by a growing community all over the world. Meanwhile, an international effort is underway to better organize this community, to favor collaborations and to prepare benchmarks of the different technologies. One can quote in particular the EMN and G-ENDEAVORS proposals recently submitted to H2020, as well as the regular Muographer workshop organized in Japan.

One of the most striking features of muography lies in the diversity of applications which have been developed or are currently emerging. After a first period where these applications were essentially driven by the fertile imagination of physicists, the muography has recently started to elicit the interest of several industrials. It is therefore likely that new, unexpected applications will emerge in the coming years, the field of civil engineering being, perhaps, the most promising.

The main limitation of muography being related to the modest muon statistics, its current performance can be exclusively enhanced by increasing the collecting area and maximizing the information extracted from each detected muon, both in hardware and software. On this aspect, more precise energy measurements appear as an important goal for future R&D. At longer term, muon focusing systems or even artificial, portable sources could signal a new era for muography, though the latter may be out of reach until the next century.

References

- [1] D. Pacini, *Nuovo Cim.* VI/ 3 (1912) 93.
- [2] V. Hess, *Phys. Zeit.* 13 (1912) 1084.
- [3] A.H. Compton, *Phys. Rev.* 43 (1933) 387.
- [4] B. Rossi, *Phys. Rev.* 36 (1930) 606.
- [5] T.H. Johnson, *Phys. Rev.* 43 (1933) 834.
- [6] L. Alvarez, A.H. Compton, *Phys. Rev.* 43 (1933) 835.
- [7] C. Patrignani, et al. (Particle Data Group), *Chin. Phys. C* 40 (2016) 100001.
- [8] D. Griffiths, *Introduction To Elementary Particles*, 2nd edn, Wiley-VCH, 2008 section 94.
- [9] L. Bonechi, *Proc. XXIX International Cosmic Ray Conference, ICRC Vol. 9, 2005* p. 283.
- [10] L.N. Bogdanova, et al., *Phys. Atom. Nucl.* 69 (2006) 1293.
- [11] see e.g., L.I. Dorman, *Cosmic Rays: Variations and Space Explorations*, North-Holland, Amsterdam, 1974.
- [12] P. Adamson, et al. (MINOS collaboration), *Phys. Rev. D* 81 (2010) 012001.
- [13] D. Heck, *CORSIKA, FZKA Report (1998)* 6019.
- [14] P. Antonioli, et al., *Astropart. Phys.* 7 (1997) 357.
- [15] V.A. Kudryavtsev, *Comput. Phys. Comm.* 180 (2009) 339.
- [16] M. Tueros, S. Sciutto, *Comput. Phys. Comm.* 181 (2010) 380.
- [17] S. Agostinelli, et al., *Nucl. Instrum. Methods A* 506 (2003) 250.
- [18] S. Bouteille, et al., *Nucl. Instrum. Methods A* 834 (2016) 223.
- [19] K.N. Borozdin, et al., *Nature* 422 (2003) 277.
- [20] L.J. Schultz, et al., *IEEE Trans. Image Process.* 16 (2007) 1985.
- [21] S. Bouteille, et al., *Nucl. Instrum. Methods A* 834 (2016) 187.
- [22] I. Bodini, et al., *Meas. Sci. Technol.* 18 (2007) 3537.
- [23] A. Zenoni, et al., *arXiv:14031709*.
- [24] see e.g., F. Sauli, *Gaseous Radiation Detectors*, Cambridge University Press, 2014.
- [25] S. Procureur, R. Dupré, S. Aune, *Nucl. Instrum. Methods A* 729 (2013) 888.
- [26] E.P. George, *Commonwealth Engineer* (1955) 455.
- [27] L.W. Alvarez, et al., *Science, New Series* 167 (1970) 832.
- [28] K. Nagamine, et al., *Nucl. Instrum. Methods A* 356 (1995) 585.
- [29] H. Tanaka, et al., *Nucl. Instrum. Methods A* 507 (2003) 657.
- [30] N. Lesparre, et al., *Geophys. J. Int. A* 183 (2010) 1348.
- [31] N. Lesparre, et al., *Geosci. Instrum. Method. Data Syst.* 1 (2012) 33.

- [32] K. Jourde, et al., *Geophys. Res. Lett.* 40 (2013) 6334.
- [33] R. Nishiyama, et al., *J. Geophys. Res.* 119 (2014) 699.
- [34] K. Jourde, D. Gibert, J. Marteau, [arXiv:14115146](https://arxiv.org/abs/14115146).
- [35] M. Rosas-Carbajal, et al., *Geophys. Res. Lett.*, submitted for publication.
- [36] K. Jourde, et al., *Nat. Sci. Rep.* 6 (2016) 33406.
- [37] C. Carloganu, et al., *Geosci. Instrum. Method. Data Syst.* 2 (2013) 55.
- [38] A. Portal, et al., *Geosci. Instrum. Method. Data Syst.* 2 (2013) 47.
- [39] F. Ambrosino, et al., *J. Geophys. Res. Solid Earth* 120 (2015) 7290.
- [40] P. Noli, et al., *Ann. Geophys.* 60 (2017) S0105.
- [41] A. Anastasio, et al., *Nucl. Instrum. Methods A* 732 (2013) 423.
- [42] V. Tioukov, et al., *Ann. Geophys.* 60 (2017) S0111.
- [43] L.W. Alvarez, *Adventures of a physicist*, in: Alfred P Sloan Foundation Series, 1989.
- [44] 33rd international cosmic ray conference, Rio de Janeiro (2013).
- [45] <http://www.scanpyramids.org/>.
- [46] <http://www.hip.institute/#press>.
- [47] H. Gomez, et al., XIV International Conference on Topics in Astroparticle and Underground Physics, TAUP 2015.
- [48] <https://www.decisionsciences.com/decision-sciences-announces-the-first-operational-deployment-of-the-multi-mode-passive-detection-system-for-nuclear-and-radiological-threats-at-freeport-container-port-in-the-bahamas/>.
- [49] S. Pesente, et al., *Nucl. Instrum. Methods A* 604 (2009) 738.
- [50] V. Anghel, et al., 2012 IEEE Nuclear Science Symposium and Medical Imaging Conference Record (NSS/MIC).
- [51] G.V. Russo, et al., *JINST* 9 (2014) P11008.
- [52] K. Gnanvo, et al., *Nucl. Instrum. Methods A* 652 (2011) 16.
- [53] P. Baesso, et al., *JINST* 9 (2014) C10041.
- [54] Broad Agency Announcement No BAA12-101 for Domestic Nuclear Detection Office.
- [55] G. Jonkmans, et al., *Ann. Nucl. Energy* 53 (2013) 267.
- [56] A. Clarkson, et al., *Nucl. Instrum. Methods A* 746 (2014) 64.
- [57] F. Ambrosino, et al., *JINST* 10 (2015) T06005.
- [58] S. Chatzidakis, C.K. Choi, L.H. Tsoukalas, *Nucl. Instrum. Methods A* 828 (2016) 37.
- [59] D. Poulson, et al., *Nucl. Instrum. Methods A* 842 (2017) 48.
- [60] D. Bryman, J. Bueno, J. Jansen, *ASEG Extended Abstracts* (2015) 1.
- [61] V. Kudryavtsev, et al., *Int. J. Green. Gas Contr.* 11 (2012) 21.
- [62] K. Borozdin, et al., *Phys. Rev. Lett.* A 109 (2012) 152501.
- [63] J. Perry, et al., *J. Appl. Phys.* 113 (2013) 184909.
- [64] http://www.tepco.co.jp/nu/fukushima-np/handouts/2015/images/handouts_150319_03-j.pdf (in Japanese).